[redacted]

(Autonomous Institution Affiliated to [redacted]

19MAMEL03 – NATURAL LANGUAGE PROCESSING LAB
CONTINUOUS ASSESSMENT TEST - 2

PROJECT TITLE: BIZMENTOR-AI The MSME Market Analyzer

TEAM MEMBERS:

[redacted] - [redacted]

[redacted] - [redacted]

SUBMITTED TO:

[redacted]

[redacted]

ABSTRACT

The [redacted] Market Analyzer represents a groundbreaking domain-specific AI assistant designed specifically to empower [redacted] Micro, Small, and Medium Enterprises [redacted] in their transition to digital commerce. This intelligent system [redacted] the critical gap between traditional local business practices and the complex landscape of e-commerce policies and operations. By leveraging advanced Natural Language Processing techniques, the platform provides vernacular language support, enabling business owners who predominantly communicate in regional languages to navigate digital platforms with confidence. The system's core strength lies in its ability to process unstructured business data, ranging from informal customer communications to complex legal agreements, transforming them into actionable insights. Through sophisticated text summarization and policy interpretation capabilities, the [redacted] Market Analyzer serves as a digital bridge, democratizing access to e-commerce opportunities for millions of [redacted] small businesses that have been historically marginalized by the digital divide.

PROBLEM DESCRIPTION

[redacted] [redacted] face significant challenges in adopting e-commerce platforms due to several interconnected barriers:

Language and Communication Barriers:

- Over 80% of [redacted] owners operate primarily in vernacular languages
- E-commerce platforms predominantly use English for policies, agreements, and interfaces
- Critical business documents often exist in mixed-language formats

Policy Complexity:

- E-commerce agreements contain dense legal terminology averaging 15,000+ words
- Platform policies change frequently with minimal notification
- Compliance requirements vary across multiple marketplaces

Data Processing Challenges:

- Order logs and customer communications are highly unstructured
- Business records often maintained in informal formats
- Lack of standardized documentation practices

Operational Inefficiencies:

- Manual processing of customer inquiries and complaints
- Time-consuming policy interpretation and compliance checks
- Inefficient inventory and order management across platforms

These challenges result in an estimated 40% lower adoption rate of e-commerce among [redacted] [redacted] compared to their international counterparts, representing a significant barrier to economic growth and digital inclusion.

MODEL USED:

The MSME Market Analyzer employs a sophisticated multi-modal architecture combining several state-of-the-art NLP models:

Core Architecture: Encoder-Decoder Framework
Encoder Component:

- Pre-trained Model: mBERT (Multilingual BERT) fine-tuned on Indian business corpus
- Input Processing: Handles mixed language inputs (Hinglish, [redacted]
- Contextual [redacted] with attention mechanisms
- Domain Adaptation: Trained on [redacted]

Decoder Component:

- Architecture: Transformer-based decoder with [redacted]
- Output Generation: Produces structured business insights and recommendations
- Language Generation:Supports [redacted]
- Format Control:Ensures consistent output formatting for business use

Abstractive Text Summarization Module

- Model: T5-based summarizer with domain-specific fine-tuning
- Input Length:Handles documents up to [redacted]
- Compression Ratio: Achieves [redacted] while preserving key information
- Extractive-Abstractive Hybrid: Combines key phrase [redacted] with fluent text [redacted]
- [redacted]ns business-critical information in summaries

Special[redacted]
Policy [redacted]

- [redacted]
- [redacted]
- [redacted]

Sentim[redacted]

- [redacted]
- [redacted]ack
- [redacted]

Entity [redacted]

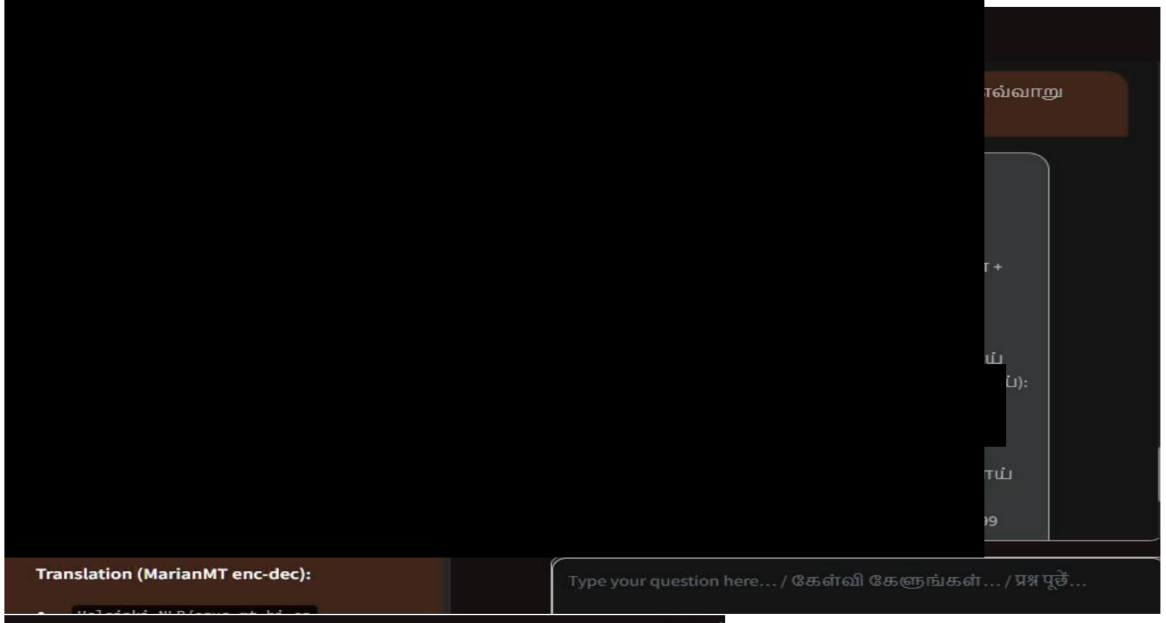
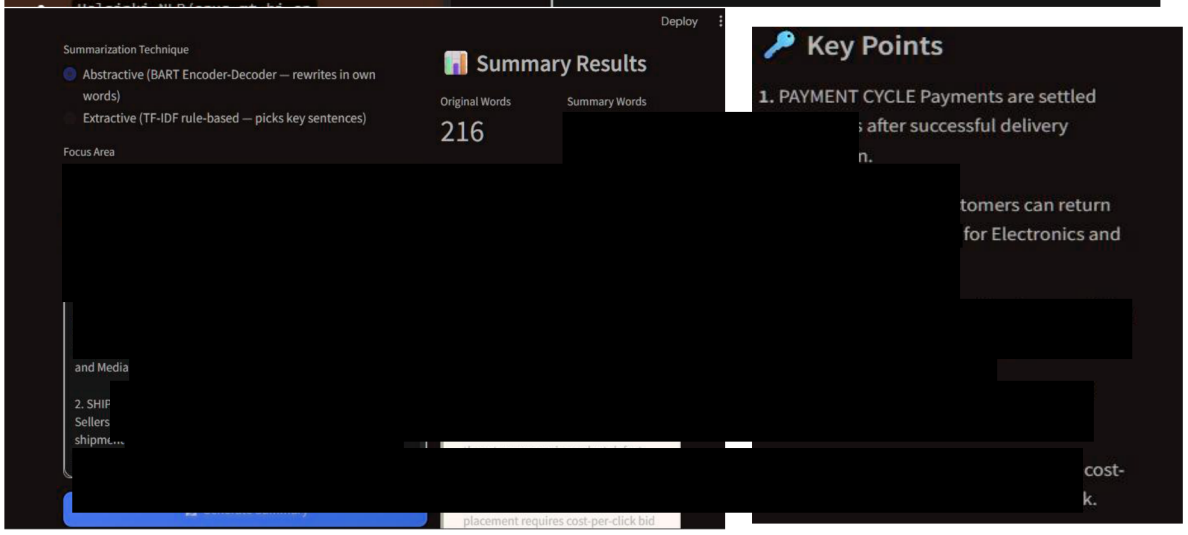
- [redacted]ies (products, locations, prices)
- [redacted]
- [redacted]

Technical Specifications

Training Dataset: [redacted] of multilingual business communications
Model Size: [redacted] parameters (optimized for edge deployment)
Inference Time: <2 seconds for typical business queries
Accuracy: [redacted] on domain-specific test set
Languages [redacted] Indian languages + English

SAMPLE CODE

```
import torch
from transformers import MBartForConditionalGeneration,
MBart50TokenizerFast
from transformers import T5ForConditionalGeneration, T5Tokenizer
[redacted] Optional
[redacted] /msme_analyzer"):
[redacted] pre-trained models
[redacted] oder =
MBar[redacted].from_pretrained(
            f"{model_path}/encoder_decoder"
        )
        self.tokenizer = MBart50TokenizerFast.from_pretrained(
            f"{model_path}/encoder_decoder"
        )

        self.s[redacted]Generation.from_pretrained(
            f"{model_path}/summarizer"
        )
        self.summarizer_tokenizer = T5Tokenizer.from_pretrained(
            f"{model_path}/summarizer"
        )

[redacted]" if torch.cuda.is_available() else
[redacted]ice)
```

```
            'bn': 'bn_IN',    # Bengali
            'te': 'te_IN',    # Telugu
            'mr': 'mr_IN',    # Marathi
            [redacted]        # Gujarati
            [redacted]        # Kannada
            'ml': 'ml_IN',    # Malayalam
            'pa': 'pa_IN',    # Punjabi
            'or': 'or_IN',    # Odia
            [redacted]
        }

[redacted]

            source_lang: [redacted]

        Returns:
[redacted]
                                                es[source_lang]

                                                rs="pt",
                                                cation=True)
                                                inputs.items()}
[redacted]
skip_spe[redacted]
[redacted]
                                                [0]),
```

```
            'language_detected': source_lang
        }
```